CISCO PACKET TRACER PROJECT

Enterprise Campus Network Design & Implementation Report

Three-tier design, configuration and validation of a two-building campus

18 configured devices | OSPF, HSRPv2, LACP, DHCP, PAT, DAI

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1. Executive Summary

Enterprise three-tier campus network

This report documents the design, configuration and validation of a three-tier campus network for a mid-sized, two-building enterprise, built in Cisco Packet Tracer. Two Layer 3 core switches connect a pair of distribution switches in each building and a customer-edge (CE) router with two ISP uplinks. The design removes single points of failure inside the campus, keeps user, voice, server, Wi-Fi and management traffic in separate VLANs, and applies the same security baseline to every device.

Item	Summary
Devices configured	18: one CE router, two ISP routers (Cisco 2911), two core switches and four distribution switches (Catalyst 3650), and nine access switches (Catalyst 2960). The lab also contains a wireless LAN controller with two lightweight access points, wired and wireless end devices, and five servers (DNS, email, NTP, Syslog, web).
Buildings	Building A (Headquarters): DSW1, DSW2, ASW1 to ASW4 (PCs, phones, printers, HR, Wi-Fi). Building B (Production site): DSW3, DSW4, ASW5 to ASW9 (servers, management Wi-Fi).
Redundancy	Dual core, dual distribution per building, HSRPv2 with per-VLAN load sharing, Rapid-PVST+ roots aligned with the active gateway, LACP bundles, two ISP links with a floating default route
Routing	OSPFv2 area 0 on the CE router, core and distribution switches. OSPFv3 on the infrastructure links. The CE router originates the default route.
Edge	PAT (NAT overload) to the ISP, an inbound ACL on both WAN links, a static NAT entry that publishes the web server, DHCP server for all user VLANs on the CE router
Switch security	Sticky port security, BPDU guard, DHCP snooping (Building A and the Wi-Fi VLAN in Building B) and Dynamic ARP Inspection, unused VLAN 999 as native VLAN, unused ports shut down
Management	SSH version 2 only, VTY access limited to 10.0.0.0/27, one management VLAN (99) per building, Syslog, NTP and DNS
Validation	Ten test cases, all passed (Section 6). A final audit of the configurations found a few gaps; they are listed with fixes in Section 8.2.

Key technical highlights

- **High availability:** HSRPv2 gateway pairs share the load by VLAN, LACP bundles join the core pair and each distribution pair, and a floating static default route provides a backup ISP.
- **Dynamic routing and segmentation:** single-area OSPF carries all campus routes, while one VLAN and one subnet per function keep traffic types apart.
- **Layer 2 defense in depth:** port security and BPDU guard protect every access port, and DHCP snooping and DAI protect the user and Wi-Fi VLANs.
- **Edge connectivity:** PAT gives private hosts Internet access through the CE router.

2. Business & Technical Requirements

The network was designed around the following business needs: application availability, separation of voice traffic, restricted access between departments, safe guest Wi-Fi and controlled administration. The table shows how each need was implemented and its status in the final configuration.

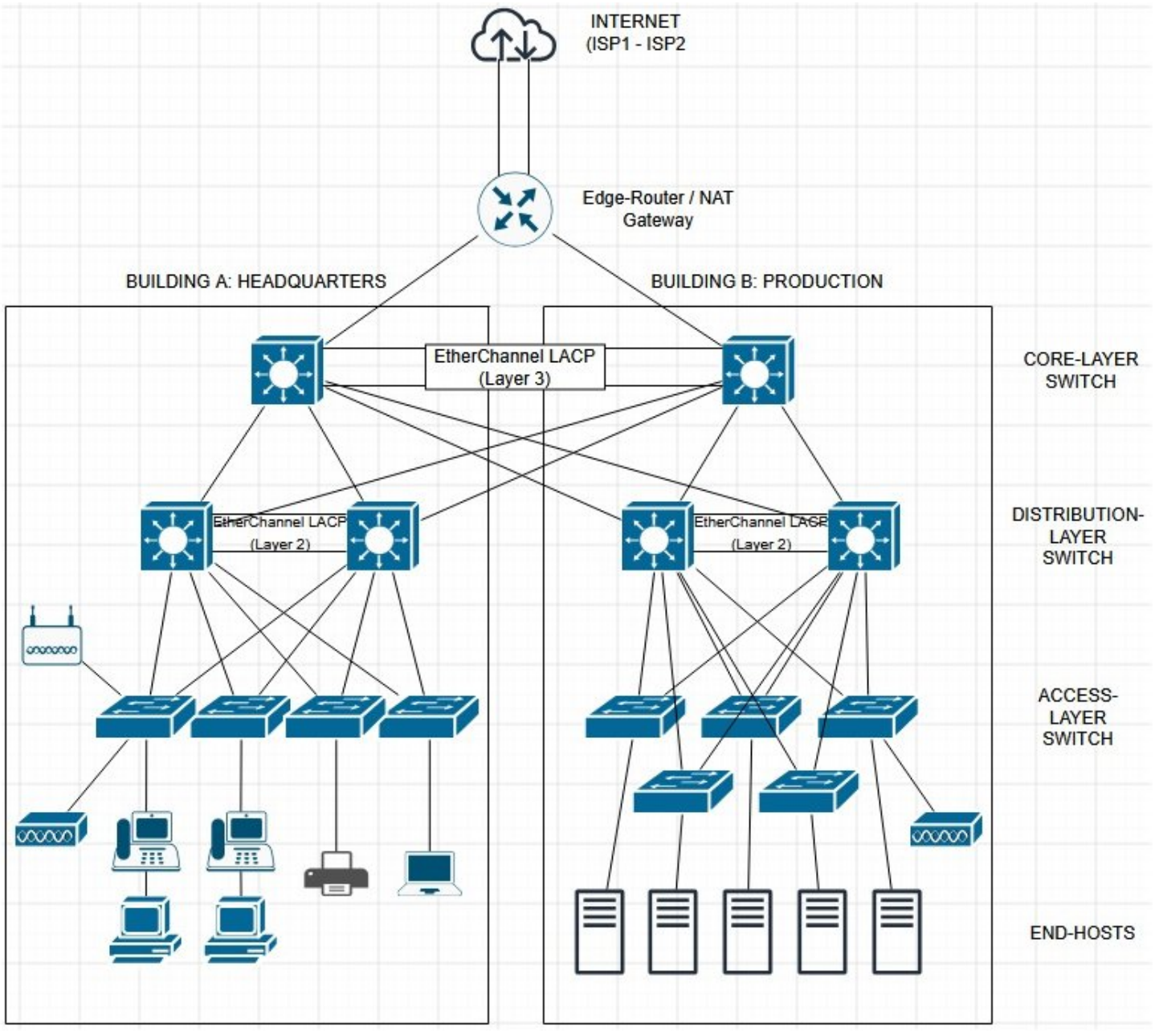
Requirement	Business reason	Technical implementation	Status
High availability	Business-critical applications stay online during a link or device failure	HSRPv2 gateway pairs with per-VLAN load sharing, Rapid-PVST+ roots aligned with the active gateway, LACP bundles, dual core uplinks, two ISP links with a floating default route	Implemented. No HSRP tracking.
Voice VLAN	Keeps IP-telephony traffic apart from data	Voice VLAN 20 (10.2.0.0/24) on the Building A ports that serve a phone and a PC (ASW1, ASW2)	Implemented as VLAN separation. QoS marking is not configured.
Network segmentation	Limits lateral movement between departments	One VLAN and subnet per function, routed only at the distribution layer	Partial. VLANs are in place; inter-VLAN ACLs are not applied.
Guest Wi-Fi	Visitors reach the Internet without reaching internal systems	Dedicated Wi-Fi VLAN 98 in Building A, guest SSID on the wireless controller, DHCP, PAT to the ISP	Internet access only. Isolation is not enforced, by decision (accepted risk in this lab).
Secure management	Only authorized administrators can configure devices	SSH v2 only, local accounts with hashed secrets, VTY ACL 10.0.0.0/27, management VLAN 99, Syslog and NTP	Implemented
Core services	Name resolution, time, logging and addressing	DNS 10.5.0.4, NTP 10.7.0.4 and Syslog 10.8.0.4 on server VLANs; DHCP on the CE router	Implemented

3. Network Topology & Architecture

3.1 High-level topology

The network follows the classic Core–Distribution–Access model. Building A is the headquarters and Building B is the production site. Every distribution switch has one routed link to each core switch, and every access switch has one trunk to each distribution switch of its building.

Layer	Devices	Role
Edge	CE-Router, ISP-1, ISP-2 (Cisco 2911; the ISP routers simulate the Internet)	CE: DHCP server, PAT, default routes and the two WAN links
Core	CSW1, CSW2 (Catalyst 3650)	Routed backbone. Connects the CE router and the four distribution switches. Routed LACP bundle (Po1) between the two cores.
Distribution	DSW1, DSW2 (Building A); DSW3, DSW4 (Building B) (Catalyst 3650)	Layer 2 and Layer 3 boundary: SVIs, HSRP gateways, DHCP relay and STP roots
Access	ASW1 to ASW4 (Building A); ASW5 to ASW9 (Building B) (Catalyst 2960)	Port security, VLAN access, voice VLAN, DHCP snooping and DAI
Wireless	WLC1, LWAP1, LWAP2	Controller at 10.0.0.8 on ASW1 Fa0/3. Access points connect to trunk ports (ASW1 Fa0/2 and ASW9 Fa0/2) carrying VLANs 98 and 99.



3.2 Physical topology

R1	G0/0	CSW1 - G1/0/7
	G0/1	CSW2 - G1/0/7
	S0/3/0	ISP-1 - S0/3/0
	S0/3/1	ISP-2 - S0/3/0
CSW1	G1/0/7	R1 - G0/0
PortChannel1	G1/0/1	CSW2 - G1/0/1
	G1/0/2	CSW2 - G1/0/2
	G1/0/3	DSW1 - G1/0/7
	G1/0/4	DSW2 - G1/0/7
	G1/0/5	DSW3 - G1/0/8
	G1/0/6	DSW4 - G1/0/8
CSW2	G1/0/7	R1 - G0/1
PortChannel1	G1/0/1	CSW1 - G1/0/1
	G1/0/2	CSW1 - G1/0/2
	G1/0/3	DSW1 - G1/0/8
	G1/0/4	DSW2 - G1/0/8
	G1/0/5	DSW3 - G1/0/9
	G1/0/6	DSW4 - G1/0/9

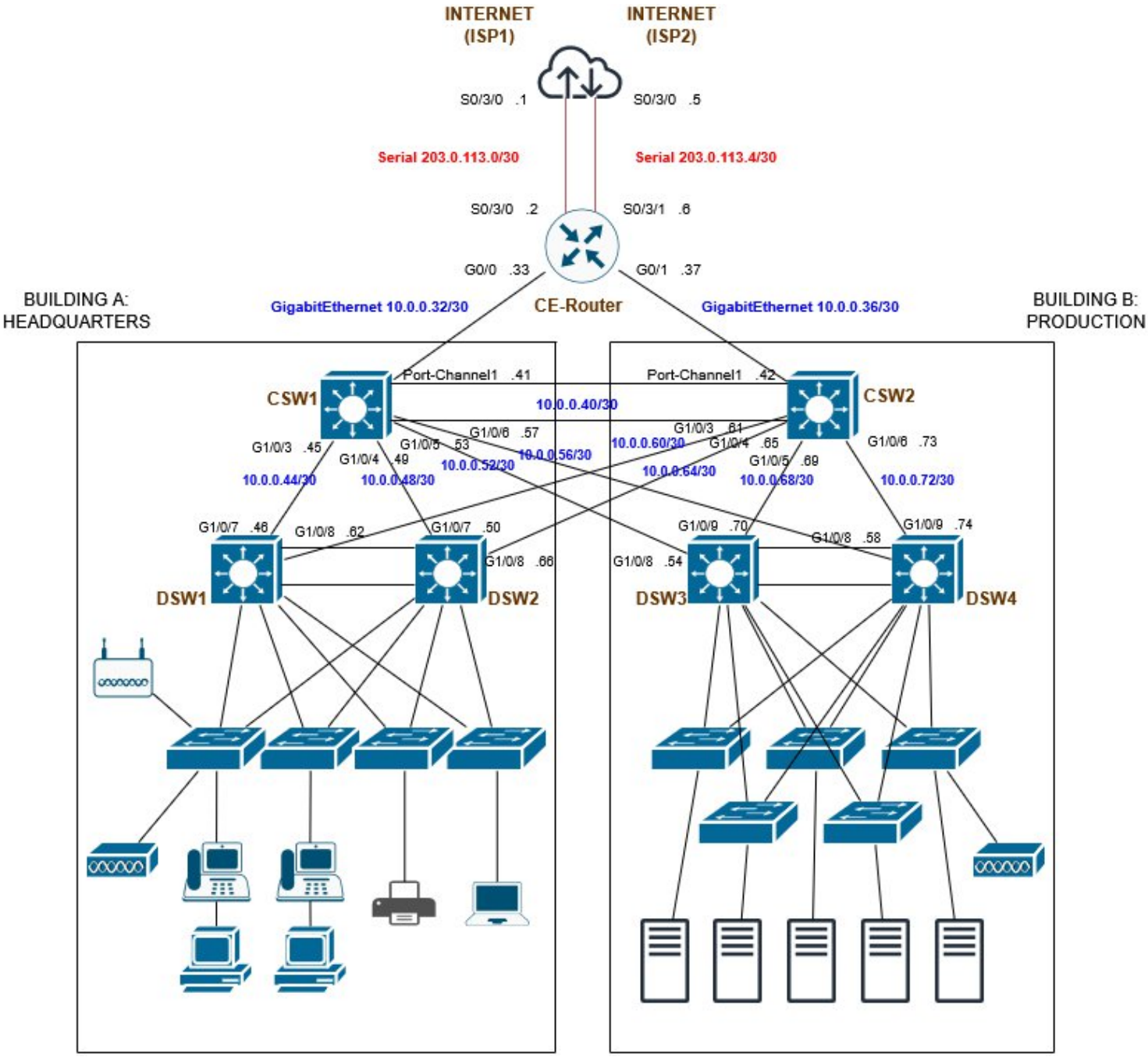
DEVICE	MODEL
CE-Router (x1)	Cisco 2911
Core Switches (x2)	Catalyst 3650 24 PS
Distribution Switches (x4)	Catalyst 3650 24 PS
Access Switches (x9)	Catalyst 2960

DSW1	G1/0/3	ASW1 - G0/1
	G1/0/4	ASW2 - G0/1
	G1/0/5	ASW3 - G0/1
	G1/0/6	ASW4 - G0/1
PortChannel1	G1/0/1	DSW2 - G1/0/1
	G1/0/2	DSW2 - G1/0/2
	G1/0/7	CSW1 - G1/0/3
	G1/0/8	CSW2 - G1/0/3
DSW2	G1/0/3	ASW1 - G0/2
	G1/0/4	ASW2 - G0/2
	G1/0/5	ASW3 - G0/2
	G1/0/6	ASW4 - G0/2
PortChannel1	G1/0/1	DSW1 - G1/0/1
	G1/0/2	DSW1 - G1/0/2
	G1/0/7	CSW1 - G1/0/4
	G1/0/8	CSW2 - G1/0/4
DSW3	G1/0/3	ASW5 - G0/1
	G1/0/4	ASW6 - G0/1
	G1/0/5	ASW7 - G0/1
	G1/0/6	ASW8 - G0/1
	G1/0/7	ASW9 - G0/1
PortChannel1	G1/0/1	DSW4 - G1/0/1
	G1/0/2	DSW4 - G1/0/2
	G1/0/8	CSW1 - G1/0/5
	G1/0/9	CSW2 - G1/0/5
DSW4	G1/0/3	ASW5 - G0/2
	G1/0/4	ASW6 - G0/2
	G1/0/5	ASW7 - G0/2
	G1/0/6	ASW8 - G0/2
	G1/0/7	ASW9 - G0/2
PortChannel1	G1/0/1	DSW3 - G1/0/1
	G1/0/2	DSW3 - G1/0/2
	G1/0/8	CSW1 - G1/0/6
	G1/0/9	CSW2 - G1/0/6

ASW1	F0/1	PC1 & Phone1
	F0/2	LWAP1
	F0/3	WLC1
	G0/1	DSW1 - G1/0/3
	G0/2	DSW2 - G1/0/3
ASW2	F0/1	PC2 & Phone2
	G0/1	DSW1 - G1/0/4
	G0/2	DSW2 - G1/0/4
ASW3	F0/1	Printer1
	G0/1	DSW1 - G1/0/5
	G0/2	DSW2 - G1/0/5
ASW4	F0/1	HR
	G0/1	DSW1 - G1/0/6
	G0/2	DSW2 - G1/0/6
ASW5	F0/1	DNS Server
	G0/1	DSW3 - G1/0/3
	G0/2	DSW4 - G1/0/3
ASW6	F0/1	Email Server
	G0/1	DSW3 - G1/0/4
	G0/2	DSW4 - G1/0/4
ASW7	F0/1	NTP Server
	G0/1	DSW3 - G1/0/5
	G0/2	DSW4 - G1/0/5
ASW8	F0/1	Syslog Server
	G0/1	DSW3 - G1/0/6
	G0/2	DSW4 - G1/0/6
ASW9	F0/1	Web Server
	F0/2	LWAP2
	G0/1	DSW3 - G1/0/7
	G0/2	DSW4 - G1/0/7

Link	Endpoint A	Endpoint B	Type
CE to ISP	CE-Router S0/3/0 and S0/3/1	ISP-1 S0/3/0 and ISP-2 S0/3/0	Serial WAN
ISP to ISP	ISP-1 S0/3/1	ISP-2 S0/3/1	Serial, IPv6 routing only
CE to core	CE-Router G0/0 and G0/1	CSW1 G1/0/7 and CSW2 G1/0/7	Routed /30
Core to core	CSW1 G1/0/1–2	CSW2 G1/0/1–2	Routed LACP (Po1)
Core to DSW1, DSW2	CSW1 G1/0/3 and G1/0/4; CSW2 G1/0/3 and G1/0/4	DSW1 G1/0/7 and G1/0/8; DSW2 G1/0/7 and G1/0/8	Routed /30
Core to DSW3, DSW4	CSW1 G1/0/5 and G1/0/6; CSW2 G1/0/5 and G1/0/6	DSW3 G1/0/8 and G1/0/9; DSW4 G1/0/8 and G1/0/9	Routed /30
DSW pairs	DSW1 G1/0/1–2; DSW3 G1/0/1–2	DSW2 G1/0/1–2; DSW4 G1/0/1–2	Trunk LACP (Po1)
DSW to access, Building A	DSW1 and DSW2 G1/0/3 to G1/0/6	ASW1 to ASW4 G0/1 and G0/2	Trunk
DSW to access, Building B	DSW3 and DSW4 G1/0/3 to G1/0/7	ASW5 to ASW9 G0/1 and G0/2	Trunk

3.3 Logical topology



- **Layer 2 domains stay inside a building.** The core links are routed, so VLAN 99 exists separately in Building A and Building B, with different subnets. VLAN 98 (guest Wi-Fi) exists in Building A only.
- **Layer 3 starts at the distribution layer.** Each VLAN has an SVI on both switches of its building and an HSRP virtual IP used as the host default gateway.
- **One OSPF area.** The CE router, both core switches and the four distribution switches run OSPF area 0, and the CE router originates the default route.

3.4 VLAN segmentation & IP addressing schema

3.4.1 VLANs and subnets

Each distribution switch uses the next addresses of every subnet: the first switch of a pair takes .2 and the second takes .3 (management VLANs: .2/.3 in Building A and .18/.19 in Building B). Servers use .4 in their subnet.

VLAN	Purpose	Subnet	HSRP VIP	Active gateway	Building
10	PCs (Sales)	10.1.0.0/24	10.1.0.1	DSW1	A
20	Voice (phones)	10.2.0.0/24	10.2.0.1	DSW1	A
30	Printers	10.3.0.0/24	10.3.0.1	DSW1	A
40	HR	10.4.0.0/24	10.4.0.1	DSW2	A
98	Wi-Fi (guest)	10.10.0.0/24	10.10.0.1	DSW2	A
99	Management	10.0.0.0/28	10.0.0.1	DSW2	A
50	DNS server	10.5.0.0/24	10.5.0.1	DSW3	B
60	Email server	10.6.0.0/24	10.6.0.1	DSW3	B
70	NTP server	10.7.0.0/24	10.7.0.1	DSW3	B
80	Syslog server	10.8.0.0/24	10.8.0.1	DSW3	B
90	Web server	10.9.0.0/24	10.9.0.1	DSW4	B
99	Management	10.0.0.16/28	10.0.0.17	DSW4	B
999	Native VLAN, unused	–	–	–	A and B

3.4.2 Point-to-point links

Link	Subnet	Endpoint A	Endpoint B
CE – ISP-1	203.0.113.0/30	CE S0/3/0 .2	ISP-1 S0/3/0 .1
CE – ISP-2	203.0.113.4/30	CE S0/3/1 .6	ISP-2 S0/3/0 .5
ISP-1 – ISP-2	203.0.113.8/30	ISP-1 S0/3/1 .9	ISP-2 S0/3/1 .10
CE – CSW1	10.0.0.32/30	CE G0/0 .33	CSW1 G1/0/7 .34
CE – CSW2	10.0.0.36/30	CE G0/1 .37	CSW2 G1/0/7 .38
CSW1 – CSW2	10.0.0.40/30	CSW1 Po1 .41	CSW2 Po1 .42
CSW1 – DSW1	10.0.0.44/30	CSW1 G1/0/3 .45	DSW1 G1/0/7 .46
CSW1 – DSW2	10.0.0.48/30	CSW1 G1/0/4 .49	DSW2 G1/0/7 .50
CSW1 – DSW3	10.0.0.52/30	CSW1 G1/0/5 .53	DSW3 G1/0/8 .54
CSW1 – DSW4	10.0.0.56/30	CSW1 G1/0/6 .57	DSW4 G1/0/8 .58
CSW2 – DSW1	10.0.0.60/30	CSW2 G1/0/3 .61	DSW1 G1/0/8 .62
CSW2 – DSW2	10.0.0.64/30	CSW2 G1/0/4 .65	DSW2 G1/0/8 .66
CSW2 – DSW3	10.0.0.68/30	CSW2 G1/0/5 .69	DSW3 G1/0/9 .70
CSW2 – DSW4	10.0.0.72/30	CSW2 G1/0/6 .73	DSW4 G1/0/9 .74

3.4.3 Management network and loopbacks

Network	Subnet	Addresses
Management, Building A	10.0.0.0/28	VIP .1, DSW1 .2, DSW2 .3, ASW1 .4, ASW2 .5, ASW3 .6, ASW4 .7, WLC1 .8
Management, Building B	10.0.0.16/28	VIP .17, DSW3 .18, DSW4 .19, ASW5 .20, ASW6 .21, ASW7 .22, ASW8 .23, ASW9 .24
Loopback0 (router IDs)	10.0.0.76 to .82 /32	CE-Router .76, CSW1 .77, CSW2 .78, DSW1 .79, DSW2 .80, DSW3 .81, DSW4 .82
Simulated Internet	8.8.8.8/32, 9.9.9.9/32	ISP-1 Loopback0 8.8.8.8, ISP-2 Loopback0 9.9.9.9

3.4.4 IPv6 transit links

IPv6 is configured on the infrastructure links only, with OSPFv3 area 0. The WAN interfaces use EUI-64 addresses in 2001:DB8:D::/64 (CE to ISP-1), 2001:DB8:C::/64 (CE to ISP-2) and 2001:DB8:F::/64 (ISP-1 to ISP-2, running EIGRP for IPv6). Host VLANs have no IPv6 addresses.

Link	Prefix	Endpoint A	Endpoint B
CE – CSW1	2001:DB8:A::/126	CE G0/0 ::0	CSW1 G1/0/7 ::3
CE – CSW2	2001:DB8:B::/126	CE G0/1 ::0	CSW2 G1/0/7 ::3
CSW1 – CSW2	2001:DB8:E1::/126	CSW1 Po1 ::0	CSW2 Po1 ::3
CSW1 – DSW1 / DSW2	2001:DB8:E2:: / E3::/126	CSW1 G1/0/3, G1/0/4 ::0	DSW1, DSW2 G1/0/7 ::3
CSW1 – DSW3 / DSW4	2001:DB8:E4:: / E5::/126	CSW1 G1/0/5, G1/0/6 ::0	DSW3, DSW4 G1/0/8 ::3
CSW2 – DSW1 / DSW2	2001:DB8:E6:: / E7::/126	CSW2 G1/0/3, G1/0/4 ::0	DSW1, DSW2 G1/0/8 ::3
CSW2 – DSW3 / DSW4	2001:DB8:E8:: / E9::/126	CSW2 G1/0/5, G1/0/6 ::0	DSW3, DSW4 G1/0/9 ::3

On every link one end uses the :: address and the other end uses ::3, the first and last addresses of the /126. The conventional host addresses would be ::1 and ::2 (see Section 8.2).

4. Design Details & Protocol Selection

4.1 Campus layering

- **Core layer:** two Catalyst 3650 switches form the Layer 3 backbone. All their links are routed, so no Spanning Tree runs across the core. They connect the CE router and the four distribution switches.
- **Distribution layer:** Catalyst 3650 switches terminate the VLANs, host the SVIs and the HSRP gateways, relay DHCP requests and hold the STP root for their VLANs.
- **Access layer:** Catalyst 2960 switches provide port security, VLAN access, the voice VLAN (802.1Q), DAI and, on the user and Wi-Fi VLANs, DHCP snooping to PCs, phones, printers, servers and the wireless access points.

4.2 High availability and redundancy

- **HSRP version 2.** Each building has an active/standby pair, with a virtual IP per VLAN (for example 10.1.0.1 for VLAN 10 and 10.0.0.1 for management). The active switch has priority 105 with preempt. The roles are split by VLAN so that both switches of a pair forward traffic:

VLANs	Active (priority 105)	Standby	Building
10, 20, 30	DSW1	DSW2	A
40, 98, 99	DSW2	DSW1	A
50, 60, 70, 80	DSW3	DSW4	B
90, 98, 99	DSW4	DSW3	B

- **Failover time.** HSRP runs with its default timers (hello 3 seconds, hold 10 seconds). A link failure that stops the active switch's hellos is therefore detected within the hold time, up to 10 seconds.
- **Rapid-PVST+.** The active gateway of each VLAN is also the STP root bridge (priority 0), so Layer 2 traffic flows toward the active gateway without crossing the inter-switch trunk.
- **LACP EtherChannel (802.3ad).** Two Gigabit links are bundled into Port-channel 1 between the core switches (routed), between DSW1 and DSW2 and between DSW3 and DSW4 (trunks). A bundle looks like a single link to Spanning Tree, so no member is blocked, and it doubles the bandwidth.
- **Core uplinks.** Each distribution switch has one routed link to each core switch, which gives two equal-cost paths. These links are individual routed /30 subnets, not bundles.

4.3 Dynamic routing (OSPF)

- **Single-area OSPF (area 0)** runs on the CE router, both core switches and the four distribution switches. The router ID of each device is its Loopback0 address.
- **Point-to-point network type** is set on the routed /30 links, which avoids the DR/BDR election and speeds up the formation of adjacencies. The /30 subnets conserve address space.
- **Passive interfaces.** The user, server and Wi-Fi SVIs and the loopbacks are passive. Their subnets are advertised without sending hello packets towards hosts.
- **OSPFv3** runs on the same infrastructure links, with the same router IDs, and the CE router originates both an IPv4 and an IPv6 default route.

4.4 Edge connectivity, NAT/PAT and ACLs

- **Two ISP links.** Serial 0/3/0 (ISP-1) is primary, with a default route of administrative distance 1 via 203.0.113.1. Serial 0/3/1 (ISP-2) is the backup, with a floating default route of distance 2 via 203.0.113.5. The same pattern applies to IPv6.
- **PAT.** Standard ACL 1 selects the private ranges (10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16) for translation with overload. The two CE links to the core are `ip nat inside`, and the two serial links are `ip nat outside`.
- **Inbound WAN filter.** The extended ACL `INTERNET_RESTRICT` is applied `in` on both serial interfaces. It permits established TCP replies to the internal range, HTTPS to the public address 203.0.113.2, ICMP echo replies and,

with an explicit final deny ip any any, nothing else. A static NAT entry (ip nat inside source static tcp 10.9.0.4 443 203.0.113.2 443) maps that address to the web server. The overload statement for PAT references Serial 0/3/1 only (Section 8.2, item 2).

- **Standard ACLs.** ACL 1 selects NAT traffic and ACL 2 restricts SSH on the VTY lines.
- **ISP routers.** ISP-1 and ISP-2 simulate the provider. Each has a loopback as an Internet destination (8.8.8.8 and 9.9.9.9) and a default route back to the CE router. Each also has a static host route to the other ISP's loopback over the link between them, so both loopbacks are reachable through either ISP. Their CE-facing serial interface carries an anti-spoofing ACL that drops packets with private source addresses (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16), so every packet must be translated by PAT before it leaves the campus. A third serial link joins the two ISPs (203.0.113.8/30); it carries IPv6 routing and the two host routes.

4.5 Wireless and management infrastructure

- **WLC and LWAP.** Lightweight access points connect to trunk ports (VLANs 98 and 99, native VLAN 99) on ASW1 and ASW9, and are managed by the controller WLC1 (10.0.0.8, ASW1 Fa0/3). The DHCP pools for the management networks carry option 43 pointing to the controller.

WLC1 configuration

Item	Setting
Management interface	10.0.0.8/28, gateway 10.0.0.1, DNS 10.5.0.4 (VLAN 99, Building A)
WLAN Management	SSID Management, VLAN 99, WPA2-PSK with AES encryption, local switching and local authentication
WLAN Guest	SSID Guest, VLAN 98, WPA2-PSK with AES encryption, local switching and local authentication
AP groups	Guest: WLAN Guest, access point LWAP1. Management: WLAN Management, access point LWAP2. This is deliberate: guests are in Building A (LWAP1), while the Network Engineer and the Administrator use the Management SSID on LWAP2. Each SSID is broadcast by one access point only.
WLC DHCP service	On, with one pool. Pool Guest-WiFi: 10.10.0.0/24, gateway 10.10.0.1, first address 10.10.0.4, up to 251 users, DNS 10.5.0.4, WLC address 10.0.0.8. Guest clients report 10.10.0.1 as their DHCP server. The management pool was removed: the Network Engineer's tablet connects through LWAP2 in Building B and leases an address in 10.0.0.16/28 (gateway 10.0.0.17) from the CE router.

With local switching, each access point places wireless clients directly on the WLAN's VLAN at its switch port. That is why the access-point trunk ports carry VLANs 98 and 99.

- **Dedicated in-band management.** Administration uses management VLAN 99 in each building (10.0.0.0/28 and 10.0.0.16/28), separate from user traffic. It is reached through the same network, so it is in-band management, protected by SSH and the VTY ACL.

5. Configuration Documentation

5.1 Configuration philosophy & standards

All devices follow one baseline to keep operation consistent, secure and easy to maintain:

- **Naming and management services:** SSH version 2 (`ip ssh version 2`), local administrator accounts with hashed secrets (type 5, MD5-based), the domain name yannickccna.com, and the central services Syslog (10.8.0.4, trap level debugging), NTP (10.7.0.4) and DNS (10.5.0.4).
- **In-band access control:** every VTY line uses `access-class 2 in` with standard ACL 2 (`permit 10.0.0.0 0.0.0.31`), so SSH is accepted only from the two management subnets (10.0.0.0/27 covers 10.0.0.0/28 and 10.0.0.16/28). Telnet is disabled with `transport input ssh`.
- **Port and VLAN hygiene:** every unused access port is shut down, VLAN 1 is shut down, and trunks use native VLAN 999 with `switchport nonegotiate` and an explicit allowed-VLAN list.
- **Unused service hardening:** the Catalyst 3650 core and distribution switches run `no profinet` to disable the industrial PROFINET service they enable by default.

5.2 Core switching layer (CSW1 / CSW2)

The core provides routed, Spanning-Tree-free transport between the CE router and the distribution layer. CSW1 is shown; CSW2 mirrors it with its own addresses (Section 3.4).

```
interface Port-channel1
 description Routed Core Inter-Switch Link
 no switchport
 ip address 10.0.0.41 255.255.255.252
 ipv6 address 2001:DB8:E1::/126
 ipv6 ospf 1 area 0
!
interface GigabitEthernet1/0/1
 no switchport
 channel-protocol lacp
 channel-group 1 mode active
!
interface GigabitEthernet1/0/3
 description Link to DSW1
 no switchport
 ip address 10.0.0.45 255.255.255.252
 ip ospf network point-to-point
 ipv6 address 2001:DB8:E2::/126
 ipv6 ospf 1 area 0
!
router ospf 1
 router-id 10.0.0.77
 passive-interface Loopback0
 network 10.0.0.77 0.0.0.0 area 0
 network 10.0.0.32 0.0.0.3 area 0
 network 10.0.0.40 0.0.0.3 area 0
 network 10.0.0.44 0.0.0.3 area 0
 network 10.0.0.48 0.0.0.3 area 0
 network 10.0.0.52 0.0.0.3 area 0
 network 10.0.0.56 0.0.0.3 area 0
!
ipv6 router ospf 1
 router-id 10.0.0.77
```

The listing is an excerpt: the other core ports (Gi1/0/2 and Gi1/0/4 to Gi1/0/7) follow the same pattern.

Key core architecture highlights

- **Routed LACP EtherChannel:** Port-channel 1 is a Layer 3 interface (`no switchport`) with its own /30 and /126 subnets, so no Layer 2 loop is possible between the cores.
- **Point-to-point adjacencies:** the routed links to the CE router and the distribution switches use `ip ospf network point-to-point`, which removes the DR/BDR election. Port-channel 1 keeps the default broadcast type (Section 8.2).

5.3 Distribution switching layer (DSW1 to DSW4)

The distribution layer is the Layer 2 and Layer 3 boundary. It runs the HSRP gateways, DHCP relay and STP root placement. DSW1 is shown as the example.

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30 priority 0
spanning-tree vlan 40,98-99 priority 4096
!
interface Vlan10
 description Sales Subnet Gateway (Active)
 ip address 10.1.0.2 255.255.255.0
 ip helper-address 10.0.0.76
 standby version 2
 standby 1 ip 10.1.0.1
 standby 1 priority 105
 standby 1 preempt
!
router ospf 1
 router-id 10.0.0.79
 passive-interface Loopback0
 passive-interface Vlan10
 passive-interface Vlan20
 passive-interface Vlan30
 passive-interface Vlan40
 passive-interface Vlan98
 network 10.0.0.79 0.0.0.0 area 0
 network 10.0.0.44 0.0.0.3 area 0
 network 10.0.0.60 0.0.0.3 area 0
 network 10.0.0.0 0.0.0.15 area 0
 network 10.1.0.0 0.0.0.255 area 0
```

The listing is an excerpt that shows VLAN 10. DSW2 has the mirror image (root and active gateway for VLANs 40, 98 and 99). DSW3 and DSW4 follow the same design for Building B:

Switch	STP root (priority 0)	HSRP active (priority 105, preempt)	DHCP relay target
DSW1	VLANs 10, 20, 30	VLANs 10, 20, 30	10.0.0.76 (CE loopback), all SVIs
DSW2	VLANs 40, 98, 99	VLANs 40, 98, 99	10.0.0.76 (CE loopback), all SVIs
DSW3	VLANs 50, 60, 70, 80	VLANs 50, 60, 70, 80	10.0.0.76 (CE loopback), VLAN 99
DSW4	VLANs 90, 99	VLANs 90, 99	10.0.0.76 (CE loopback), VLAN 99

Key distribution architecture highlights

- **Active/standby load sharing:** each switch of a pair is the HSRP active gateway and the Rapid-PVST+ root for half of its building's VLANs, so both switches carry traffic.
- **DHCP relay:** each user SVI forwards DHCP broadcasts as unicast to the DHCP server on the CE router (Section 5.5).
- **Passive SVIs:** user, server and Wi-Fi SVIs are passive in OSPF. Their subnets are advertised to the core without hello packets towards hosts.

5.4 Access switching layer (ASW1 to ASW9)

The access layer enforces the edge security policy. ASW1 is shown as the example.

```
ip arp inspection vlan 10,20,30,40,98,99
ip arp inspection validate src-mac dst-mac ip
ip dhcp snooping vlan 10,20,30,40,98,99
no ip dhcp snooping information option
ip dhcp snooping
!
interface FastEthernet0/1
 description Workstation + IP Phone (Sales Subnet)
 switchport access vlan 10
 switchport mode access
 switchport nonegotiate
```

```
 switchport voice vlan 20
 switchport port-security
 switchport port-security maximum 3
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface GigabitEthernet0/1
 description Uplink to DSW1
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40,98-99
 switchport mode trunk
 switchport nonegotiate
 ip dhcp snooping trust
 ip arp inspection trust
```

Switch	Access port	Connected device	VLAN	Management IP
ASW1	Fa0/1	Workstation and IP phone	10 (voice 20)	10.0.0.4
	Fa0/2, Fa0/3	Access point and WLC1 (trunks)	98, 99	–
ASW2	Fa0/1	Workstation and IP phone	10 (voice 20)	10.0.0.5
ASW3	Fa0/1	Printers	30	10.0.0.6
ASW4	Fa0/1	HR workstation	40	10.0.0.7
ASW5	Fa0/1	DNS server	50	10.0.0.20
ASW6	Fa0/1	Email server	60	10.0.0.21
ASW7	Fa0/1	NTP server	70	10.0.0.22
ASW8	Fa0/1	Syslog server	80	10.0.0.23
ASW9	Fa0/1; Fa0/2	Web server; access point trunk	90; 98, 99	10.0.0.24

ASW1 to ASW4 trunk to DSW1 and DSW2 and allow VLANs 10, 20, 30, 40, 98 and 99. ASW5 to ASW9 trunk to DSW3 and DSW4 and allow VLANs 50, 60, 70, 80, 90 and 99. Their default gateways are 10.0.0.1 and 10.0.0.17.

Key access architecture highlights

- **Layer 2 defense in depth:** on ASW1 to ASW4, DHCP snooping blocks rogue DHCP servers and Dynamic ARP Inspection validates ARP packets against the snooping binding table. On ASW5 to ASW9, DAI is enabled and the static server ports are trusted; ASW9 also runs DHCP snooping for VLAN 99, where the access point and the tablet lease addresses.
- **Port security and STP safeguards:** access ports learn MAC addresses with sticky learning and the `restrict` violation mode. `portfast` speeds up port start-up, and `bpduguard` shuts the port down if a switch is connected.
- **VLAN hopping mitigation:** trunks allow only the VLANs they need and use the unused native VLAN 999. The access-point trunks are the exception: they use native VLAN 99 and carry only the Wi-Fi VLANs: VLANs 98 and 99 on ASW1, and VLAN 99 on ASW9.

5.5 Edge router (CE-Router)

The CE router connects the campus to both ISPs. It is also the DHCP server for every dynamic VLAN.

```
interface GigabitEthernet0/0
 description Link to CSW1 (Building-A Core)
 ip address 10.0.0.33 255.255.255.252
 ip ospf network point-to-point
 ip nat inside
!
interface Serial0/3/0
 description Primary WAN to ISP1
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 ip access-group INTERNET_RESTRICT in
!
```

```
router ospf 1
 router-id 10.0.0.76
 default-information originate
!
ip nat inside source list 1 interface Serial0/3/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip route 0.0.0.0 0.0.0.0 203.0.113.5 2
!
ip access-list extended INTERNET_RESTRICT
 permit tcp any 10.0.0.0 0.255.255.255 established
 permit tcp any host 10.9.0.4 eq 443
 permit icmp any any echo-reply
```

The listing is an excerpt. Interface G0/1 (to CSW2) and Serial 0/3/1 (to ISP-2) mirror G0/0 and Serial 0/3/0. The DHCP pools are:

Pool	Network	Gateway	Excluded	Notes
A-Mgmt	10.0.0.0/28	10.0.0.1	10.0.0.1 – .8	Option 43: 10.0.0.8 (WLC)
B-Mgmt	10.0.0.16/28	10.0.0.17	10.0.0.17 – .24	Option 43: 10.0.0.8 (WLC)
LAN-Sales	10.1.0.0/24	10.1.0.1	10.1.0.1 – .3	VLAN 10
LAN-Phones	10.2.0.0/24	10.2.0.1	10.2.0.1 – .3	VLAN 20
LAN-Printers	10.3.0.0/24	10.3.0.1	10.3.0.1 – .3	VLAN 30
LAN-HR	10.4.0.0/24	10.4.0.1	10.4.0.1 – .3	VLAN 40
A-WiFi	10.10.0.0/24	10.10.0.1	First three addresses	VLAN 98 (Building A)

All pools hand out DNS server 10.5.0.4 and the domain yannickccna.com. The server VLANs (50 to 90) use static addresses.

5.6 ISP routers (simulated Internet)

ISP-1 and ISP-2 are outside the enterprise baseline. They have no SSH, Syslog or NTP configuration and their VTY lines use `login` without a password. ISP-1 is shown:

```
interface Loopback0
 ip address 8.8.8.8 255.255.255.255
!
interface Serial0/3/0
 ip address 203.0.113.1 255.255.255.252
 ip access-group 1 in
 ipv6 address 2001:DB8:D::/64 eui-64
 ipv6 eigrp 1
!
interface Serial0/3/1
 ip address 203.0.113.9 255.255.255.252
 ipv6 address 2001:DB8:F::/64 eui-64
 ipv6 eigrp 1
!
ip route 0.0.0.0 0.0.0.0 203.0.113.2
access-list 1 deny 10.0.0.0 0.255.255.255
access-list 1 deny 172.16.0.0 0.15.255.255
access-list 1 deny 192.168.0.0 0.0.255.255
access-list 1 permit any
```

Router	Loopback0	S0/3/0 (to CE)	S0/3/1 (to other ISP)	Default route
ISP-1	8.8.8.8	203.0.113.1/30	203.0.113.9/30	Via 203.0.113.2 (CE)
ISP-2	9.9.9.9	203.0.113.5/30	203.0.113.10/30	Via 203.0.113.6 (CE), distance 2

The ACL on the CE-facing serial interface behaves like a provider's anti-spoofing filter. The link between the two ISPs carries EIGRP for IPv6 and, for IPv4, one static host route on each router: ISP-1 reaches 9.9.9.9/32 via 203.0.113.10 and ISP-2 reaches 8.8.8.8/32 via 203.0.113.9.

5.7 Device management & IP summary table

Device	Role	Management IP	Routing / ID	Key configured roles
CE-Router	Edge	Loopback 10.0.0.76	OSPFv2/v3 (10.0.0.76)	DHCP server, PAT, ISP failover, default route origination
ISP-1, ISP-2	Simulated Internet	Loopback 8.8.8.8, 9.9.9.9	Static default route; EIGRP for IPv6	Anti-spoofing ACL on the CE-facing serial; outside the enterprise baseline
CSW1	Core	Loopback 10.0.0.77	OSPFv2/v3 (10.0.0.77)	Routed core, routed Po1 to CSW2
CSW2	Core	Loopback 10.0.0.78	OSPFv2/v3 (10.0.0.78)	Routed core, routed Po1 to CSW1
DSW1	Distribution A	10.0.0.2/28 (VLAN 99)	OSPFv2/v3 (10.0.0.79)	HSRP active and STP root for VLANs 10, 20, 30; DHCP relay
DSW2	Distribution A	10.0.0.3/28 (VLAN 99)	OSPFv2/v3 (10.0.0.80)	HSRP active and STP root for VLANs 40, 98, 99; DHCP relay
DSW3	Distribution B	10.0.0.18/28 (VLAN 99)	OSPFv2/v3 (10.0.0.81)	HSRP active and STP root for VLANs 50 to 80; DHCP relay (98, 99)
DSW4	Distribution B	10.0.0.19/28 (VLAN 99)	OSPFv2/v3 (10.0.0.82)	HSRP active and STP root for VLANs 90, 98, 99
WLC1	Wireless controller	10.0.0.8/28 (VLAN 99)	–	WLANs Management (VLAN 99) and Guest (VLAN 98), local switching, two AP groups, guest DHCP pool
ASW1 to ASW4	Access A	10.0.0.4 to .7 (VLAN 99)	Default gateway 10.0.0.1	DAI, DHCP snooping, port security; voice VLAN on ASW1 and ASW2
ASW5 to ASW9	Access B	10.0.0.20 to .24 (VLAN 99)	Default gateway 10.0.0.17	DAI with trusted server ports, port security; DHCP snooping on ASW9 (VLAN 99)

6. Test & Validation

6.1 Test strategy & execution methodology

A structured test plan checked the network in five areas. Each test lists the command used and the expected result.

- **Layer 2 connectivity and security:** VLAN isolation, trunking, Rapid-PVST+ root placement, voice VLAN, DHCP snooping, DAI and port security.
- **First-hop redundancy:** HSRPv2 states, virtual IP gateways, failover and preemption.
- **Core and distribution routing:** OSPF adjacencies, passive interfaces and the routing table.
- **Services and management:** DHCP relay, DNS, NTP, Syslog and SSH access control.
- **Internet access:** PAT and ISP failover.

6.2 Test plan & execution results

ID	Objective	Procedure and commands	Expected result	Status
TC-01	Loop prevention and root bridge placement	`ASW1# show spanning-tree vlan 10` and `DSW1# show spanning-tree vlan 10`	DSW1 is the root bridge for VLAN 10 (bridge priority 10: 0 plus the VLAN ID). On ASW1, Gi0/1 (to DSW1) is the root port, forwarding, and Gi0/2 is the alternate port, blocking.	PASS
TC-02	Port security and rogue device mitigation	Connect devices with new MAC addresses to Fa0/1 on ASW1 until the maximum of 3 learned addresses is exceeded; then `ASW1# show port-security interface fastEthernet 0/1`	Addresses up to the maximum are learned and kept as sticky entries. A further address increments the violation counter, its traffic is dropped (restrict mode) and a PSECURE_VIOLATION message is logged. Allowed devices keep working.	PASS
TC-03	HSRP failover and preemption	`DSW1# show standby brief` and `DSW2# show standby brief`; then `DSW1(config)# interface vlan 10` and `shutdown`, with a continuous ping from a Sales host to the CE router loopback 10.0.0.76; finally `no shutdown`	DSW1 is Active for VLANs 10, 20, 30 and Standby for VLANs 40, 98 and 99. After the shutdown DSW2 becomes Active for VLAN 10 and the ping resumes after a short interruption. After `no shutdown` DSW1 preempts and becomes Active again.	PASS
TC-04	OSPF adjacencies and routes	`CSW1# show ip ospf neighbor`, `show ipv6 ospf neighbor`, `show ip route ospf`	CSW1 has six neighbors: DSW1 to DSW4 (Gi1/0/3 to Gi1/0/6), the CE router (Gi1/0/7) and CSW2 (Po1). Point-to-point links show FULL/ -. The table holds all VLAN subnets (10.1.0.0/24, 10.2.0.0/24 and so on).	PASS
TC-05	DHCP relay and SSH restriction	`ASW1# show ip dhcp snooping binding`; `DSW1# show running-config | include ip helper`; `ASW1# ssh -l yannick 10.0.0.2`; the same SSH attempt from a VLAN 10 host	Workstations lease addresses from the CE router through the SVI helper address. SSH from the management subnets (10.0.0.0/27) succeeds. SSH from VLAN 10 (10.1.0.0/24) is refused by ACL 2.	PASS
TC-06	Internet access, PAT and ISP failover	From a PC ping 8.8.8.8, then `CE-Router# show ip nat translations`; shut down Serial 0/3/0, then `show ip route` and ping 9.9.9.9	The ping succeeds and the inside address is translated to the exit interface address. After the primary link goes down the default route moves to 203.0.113.5 and ISP-2 (9.9.9.9) is reachable.	PASS
TC-07	Management security	SSH to CE-Router (10.0.0.76), CSW1 (10.0.0.77), DSW1 (10.0.0.79) and ASW9 (10.0.0.24) from the Network Engineer host and from PC1 (VLAN 10); `show`	SSH from the management host succeeds on all four devices. From PC1 it is refused on all four and the deny counter of ACL 2 increases.	PASS

ID	Objective	Procedure and commands	Expected result	Status
		`access-lists 2` on each		
TC-08	Syslog	`CSW2(config)# interface g1/0/3` and `shutdown`; check the log on the Syslog server (10.8.0.4)	The adjacency-change message reaches the Syslog server.	PASS
TC-09	Voice VLAN	`ASW1# show interfaces fa0/1 switchport`; check the phone's address	Access VLAN 10 and voice VLAN 20; the phone leases an address in 10.2.0.0/24.	PASS
TC-10	DNS and NTP	`nslookup csw2` from PC2; `ASW3# show ntp status`	The name resolves through 10.5.0.4. ASW3 is synchronized to 10.7.0.4.	PASS

Recorded results

ID	Observed result
TC-01	VLAN 10: root ID priority 10 at 0001.C98C.018B, and DSW1 reports "This bridge is the root". ASW1: Gi0/1 is Root and forwarding (cost 4), Gi0/2 is Alternate and blocking, Fa0/1 is Designated and forwarding.
TC-02	Fa0/1 is Secure-up in Restrict mode with a maximum of 3, a total of 3 and 3 sticky addresses. The rogue PC (000C.85A1.BB4C, VLAN 10) triggered PSECURE_VIOLATION and the violation count reached 11. The rogue PC only obtained the self-assigned address 169.254.187.76 (DHCP was denied).
TC-03	Before the test, DSW1 was Active (priority 105, preempt) for VLANs 10, 20, 30 and Standby for 40, 99 and 98; DSW2 was the reverse. After `shutdown` on DSW1 Vlan10, DSW2 logged Vlan10 Grp 1 Standby -> Active. The ping from a Sales host to 10.0.0.76 sent 34 packets and received 33: 1 lost (3% loss), so the gateway moved with one lost packet. After `no shutdown`, `show standby brief` shows DSW1 Active again for VLANs 10, 20 and 30 and DSW2 Standby for them, with both switches back in their original roles for VLANs 40, 99 and 98.
TC-04	CSW1 has six OSPFv2 and six OSPFv3 neighbors, all FULL. The point-to-point links show FULL/ - and Po1 (10.0.0.78) shows FULL/DR. The routing table lists the infrastructure subnets, the VLAN subnets (10.1.0.0 to 10.11.0.0 when the test ran; the Building B Wi-Fi subnet 10.11.0.0 was removed afterwards) and the default route O*E2 via the CE router (10.0.0.33).
TC-05	The snooping table has two bindings on Fa0/1: the PC (10.1.0.6) and the phone (10.2.0.8), both listed under VLAN 10. DSW1 shows six `ip helper-address` lines, all 10.0.0.76 (the CE loopback). SSH from ASW1 to DSW1 (10.0.0.2) reached the DSW1 prompt. SSH from PC1 was refused ("Connection refused by remote host").
TC-06	The PC (10.1.0.6) pinged 8.8.8.8 with 4 of 4 replies. The first reply took 3219 ms because of address and route resolution; the others took 22 to 38 ms. The NAT table translates each ICMP session from 10.1.0.6 to 203.0.113.6 and also holds a static entry 203.0.113.2:443 to 10.9.0.4:443. In the failover test the only default route was [2/0] via 203.0.113.5 and the ping to 9.9.9.9 got 4 of 4 replies.
TC-07	SSH from the Network Engineer host reached the user prompt on CE-Router, CSW1, DSW1 and ASW9. From PC1 every attempt ended with "Connection refused by remote host". On each device `show access-lists 2` shows permit 10.0.0.0 0.0.0.31 (2 matches) and deny any (16 matches).
TC-08	Shutting Gi1/0/3 on CSW2 produced OSPF and OSPFv3 adjacency-down messages on CSW2. The Syslog server received the matching message sent by DSW1 ("Nbr 10.0.0.78 on GigabitEthernet1/0/8 from FULL to DOWN") at 22:04:47.
TC-09	Fa0/1 is in static access mode with access VLAN 10 (Sales) and voice VLAN 20. The IP phone leased 10.2.0.6/24.
TC-10	`nslookup csw2` from PC2 returned 10.0.0.78 from server 10.5.0.4. ASW3 reports "Clock is synchronized, stratum 2, reference is 10.7.0.4".

6.3 Test & validation summary table

Test ID	Category	Target devices	Success criterion	Result
TC-01	Spanning Tree	ASW1, DSW1, DSW2	Root bridge on DSW1 for VLANs 10 to 30; redundant uplink blocked	PASS
TC-02	Layer 2 security	ASW1	Port security restricts extra MAC addresses; DAI and DHCP snooping active	PASS
TC-03	Gateway failover	DSW1, DSW2	HSRP fails over to DSW2 after the SVI shutdown; DSW1 preempts on recovery	PASS

Test ID	Category	Target devices	Success criterion	Result
TC-04	Dual-stack OSPF	CSW1, CSW2, DSW1 to DSW4	OSPFv2 and OSPFv3 adjacencies reach FULL; all subnets are routed	PASS
TC-05	Service relay and SSH ACL	DSW1, ASW1	DHCP requests are relayed through the SVI helper address; clients acquire leases and SSH is restricted by ACL 2	PASS
TC-06	Internet and failover	CE-Router, ISP-1, ISP-2	PAT translates inside addresses; the default route fails over to 203.0.113.5 and 9.9.9.9 is reachable	PASS
TC-07	Management security	CE-Router, CSW1, DSW1, ASW9	The VTY ACL allows SSH from 10.0.0.0/27 and refuses other sources	PASS
TC-08	Logging	CSW2, DSW1, Syslog server	Adjacency-change messages reach 10.8.0.4	PASS
TC-09	Voice VLAN	ASW1, IP phone	Voice VLAN 20 on an access port; the phone leases a 10.2.0.0/24 address	PASS
TC-10	DNS and NTP	PC2, ASW3, servers	The name resolves; the clock is synchronized to 10.7.0.4	PASS

6.4 Requirements without a test

The ten tests cover Layer 2, first-hop redundancy, routing, services, management and Internet access. Two items have no test, for the reasons below:

Area	Reason	Status
Guest isolation	Not implemented by decision: VLAN 98 is not separated from the internal subnets in this lab.	Accepted risk
DHCP in Building B Wi-Fi	VLAN 98 was removed from Building B (no SVI, no pool). Building B has only the Management SSID (VLAN 99), where the administrator gets an address from the CE router.	Not applicable

7. Key Takeaways & Lessons Learned

Building this network gave hands-on experience with hierarchical design, high availability, control-plane efficiency and layered security. The main lessons are below.

7.1 Layer 3 core & dynamic routing optimization

- **Routed EtherChannels instead of Layer 2 bundles:** making the core inter-switch link a Layer 3 port-channel removes Spanning Tree from the backbone, and traffic is shared across the member links by hashing.
- **Point-to-point OSPF network type:** `ip ospf network point-to-point` skips the DR/BDR election, so adjacencies form faster and there is less control-plane overhead. It is set on every routed link except the core port-channel, because Packet Tracer does not allow the network type on a routed port-channel; Port-channel 1 keeps the broadcast type.
- **Precise network statements and passive interfaces:** exact wildcard masks and passive SVIs keep routing neighbors out of host VLANs. The management VLAN SVIs also need to be passive.

7.2 High availability & traffic alignment

- **Aligning STP and HSRP:** DSW1 is both the HSRP active gateway (priority 105) and the Rapid-PVST+ root (priority 0) for VLANs 10, 20 and 30, so Layer 2 traffic flows straight to the active gateway. DSW2 does the same for VLANs 40, 98 and 99.
- **Preemption:** `standby preempt` lets the primary gateway take back the Active role after an outage, which keeps traffic paths predictable.
- **Detection time matters:** with default HSRP timers a failure takes up to 10 seconds to detect. Shorter timers and interface tracking are the next step.

7.3 Edge security & defense in depth

- **Layer 2 attacks:** DHCP snooping with DAI protects access ports against rogue DHCP servers and ARP poisoning. Static servers on DAI-protected VLANs need trusted ports (or ARP ACLs) on real equipment, which the server ports now have.
- **VLAN hopping and native VLAN:** an unused native VLAN (999), explicit allowed lists and `nonegotiate` on all trunks. Consistent trunk definitions at both ends also avoid native-VLAN mismatch messages.
- **Management isolation:** SSH v2 only, a VTY ACL and a dedicated management VLAN reduce the exposure of device administration.

7.4 Keeping documentation and configuration in sync

- **Audit the final configuration against the report.** A line-by-line review of the exported configurations found places where the written design was ahead of the configuration (guest ACL, NAT on both ISPs) and others where the text was inaccurate (management ACL range, EtherChannel scope, server roles).
- **Test every stated requirement.** Each requirement in Section 2 should have a test in Section 6, which would have exposed the missing guest isolation earlier.

8. Conclusion & Future Enhancements

8.1 Executive conclusion

This project designed, implemented and validated a three-tier campus network for a two-building enterprise. The design provides redundant gateways and uplinks, a single-area OSPF core with an IPv6 transit layer, a dual-ISP edge with NAT, and layered Layer 2 and management security. The test cases confirm the functional behavior of spanning tree, HSRP, OSPF, DHCP relay and the management ACL.

- **Resilient infrastructure:** HSRPv2 gateway pairs aligned with Rapid-PVST+ roots, LACP bundles and dual core uplinks.
- **Routed core:** loop-free, Spanning-Tree-free transport over Layer 3 EtherChannels and point-to-point subnets.
- **Hardened edge ports:** DHCP snooping, DAI, sticky port security, BPDU guard and an isolated native VLAN.
- **Controlled administration:** SSH v2, a VTY ACL, NTP synchronization and Syslog.

The final audit also showed where the design can be tightened. Section 8.2 lists those gaps with their fixes.

8.2 Known limitations & corrections

#	Area	Finding (from the configurations)	Impact	Fix
1	Guest and inter-VLAN isolation	No ACL is applied on any DSW SVI.	Wi-Fi VLAN 98 can reach internal subnets	Accepted for this lab. An extended ACL inbound on the Vlan98 SVIs would isolate guests
2	PAT with two ISPs	`ip nat inside source list 1 interface Serial0/3/1 overload` binds PAT to the backup link only	On real IOS, traffic leaving through ISP-1 is not translated and ISP-1 drops private sources	Accepted for this lab. Per-ISP route-maps with overload on both serials would translate traffic on either link
3	HSRP timers and tracking	Default timers (hello 3 s, hold 10 s); no interface tracking	Up to 10 s failover; the active gateway ignores uplink loss	Accepted for this lab. `standby timers`, tracking and preempt on the standby peers would shorten failover
4	OSPF on VLAN 99	The VLAN 99 SVI is not passive on DSW1 to DSW4	The two switches of a pair form an adjacency over VLAN 99. This is kept: it gives a fallback path if one switch loses its core links.	Accepted. Passive on all SVIs would remove the adjacency; OSPF authentication on VLAN 99 would protect it
5	OSPF on Po1	No point-to-point network type on the core Port-channel 1	DR/BDR election on the core link	Accepted: Packet Tracer does not allow the network type on a routed port-channel
6	Console and sessions	Console has no login; VTY idle timeout is 30 minutes	Physical access and idle sessions	Console login and a 5 to 10 minute timeout
7	Access-point trunks	ASW1 Fa0/2 and Fa0/3 and ASW9 Fa0/2 are trusted for snooping and DAI, with native VLAN 99. With local switching, Wi-Fi clients enter VLANs 98 and 99 on the ASW1 AP ports and VLAN 99 on the ASW9 AP port.	Snooping and DAI are bypassed for Wi-Fi clients	Accepted: Packet Tracer does not handle access-point trunks otherwise

#	Area	Finding (from the configurations)	Impact	Fix
8	IPv6 scope and addressing	IPv6 only on infrastructure links, using the :: and ::3 addresses of each /126	Unconventional addressing; no IPv6 for hosts	Use ::1 and ::2; extend IPv6 to host VLANs later
9	ISP failover detection	Floating static routes only	A failure beyond the first hop is not detected	Accepted for this lab. IP SLA with route tracking would detect failures beyond the first hop
10	Voice QoS	No QoS or telephony-service configuration	Voice is separated but not prioritized	`mls qos` trust and queuing; telephony-service
11	Management SSID	Both SSIDs use WPA2-PSK with short, guessable passphrases; the Management SSID places clients in VLAN 99	Anyone with the passphrase reaches VLAN 99, the VLAN allowed to SSH	Accepted for this lab. Long passphrases or WPA2-Enterprise with a RADIUS server would harden it
12	Two DHCP sources	The WLC pool Guest-WiFi and the CE pool A-WiFi both cover 10.10.0.0/24	Guest clients lease from the WLC (server 10.10.0.1), so the CE pool is unused; overlapping pools risk duplicate addresses	Remove the unused CE pool A-WiFi

8.3 Future enhancements

- **Close the remaining gaps first:** PAT on both ISPs, HSRP tracking and shorter timers, IP SLA tracking of the ISP paths, and console login with a shorter idle timeout. Guest isolation is optional for this lab.
- **Centralized AAA:** TACACS+ or RADIUS with per-role privilege levels instead of shared local accounts.
- **Voice quality:** QoS trust and queuing on the access and distribution switches, and a call-control service for the IP phones.
- **Resilient edge:** a second CE router with HSRP, and IP SLA tracking of the ISP paths.
- **Routing protection:** OSPF and OSPFv3 authentication on the infrastructure links.
- **Automation and monitoring:** Python (Netmiko) or Ansible for configuration and compliance checks, configurations in Git, and NetFlow and SNMPv3 telemetry.
- **Longer term:** 802.1X network access control and an overlay fabric such as VXLAN-EVPN as the campus grows.